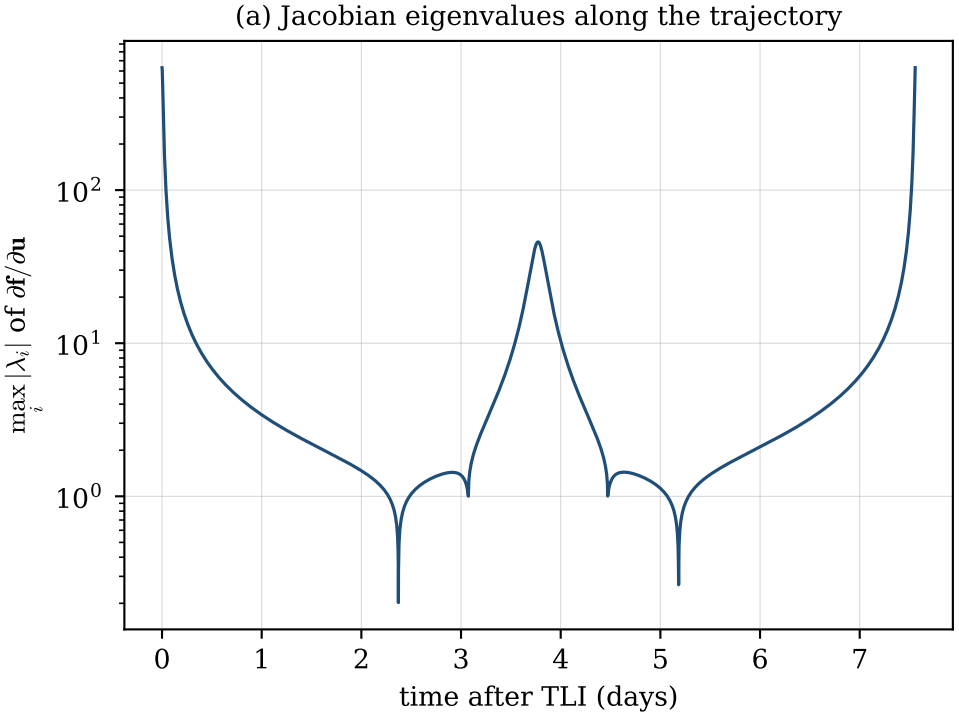
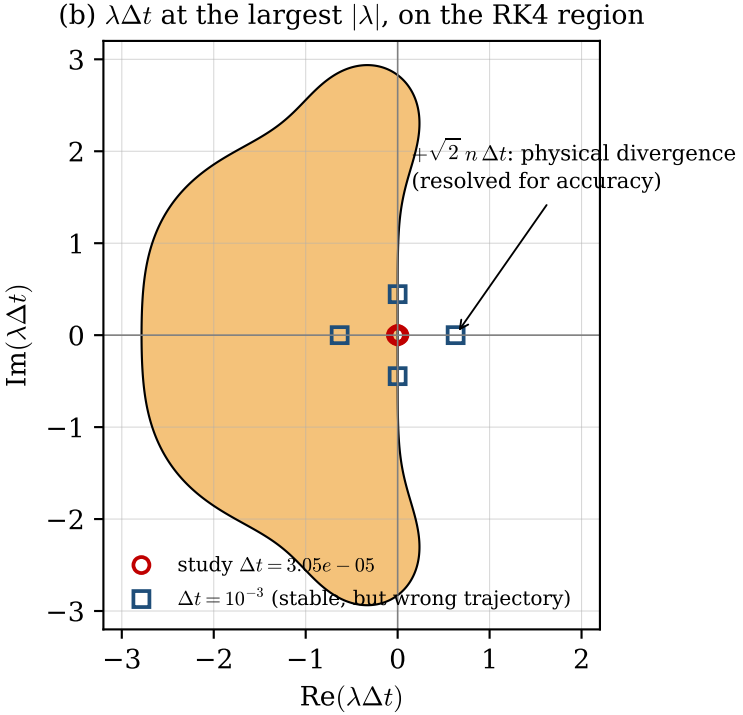

(a) Jacobian eigenvalues along the trajectory
$\max_i |\lambda_i|$ of $\partial \mathbf{f}/\partial \mathbf{u}$
10^2
10^1
10^0
0
1
2
3
4
5
6
7
time after TLI (days)


(b) $\lambda\Delta t$ at the largest $|\lambda|$, on the RK4 region
$\pm\sqrt{2}\, n\, \Delta t$: physical divergence
(resolved for accuracy)
study $\Delta t = 3.05e-05$
$\Delta t = 10^{-3}$ (stable, but wrong trajectory)
$\mathrm{Im}(\lambda\Delta t)$
$\mathrm{Re}(\lambda\Delta t)$
3
2
1
0
−1
−2
−3
−3
−2
−1
0
1
2